

CS 766 Mid-Term Report

First-Layer 3D Print Quality Detection

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1. Summary of Current Progress

Our project aims to build a classical computer vision pipeline that classifies 3D printer first-layer quality from a single top-down image into three categories: **optimal**, **under-extruded**, and **over-extruded**.

Completed

- **2D convolution** - naive loop-based, optimized stride-trick, and separable variants
- **Automatic ROI segmentation** - image segmentation with multi-strategy fallback (thresholding, HSV thresholding, inverted thresholding), morphological cleanup, and flood-fill connected-component analysis
- **Canny edge detection** - Gaussian blur, Sobel gradient operators, non-maximum suppression, double thresholding, and hysteresis edge tracking
- **Feature extraction** - fill density, gradient direction entropy, and gradient magnitude statistics
- **Nearest-centroid distance classifier** - per-class feature-space distance metric with distance weights
- **Evaluation framework** - confusion matrix, precision/recall/F1 reporting
- **Dataset** - 65 manually curated first-layer images (24 optimal, 20 under-extruded, 21 over-extruded)
- **Project website** - draft complete

In Progress / Remaining

- Validation of classifier weights on held-out data (new images being collected)
 - Texture-based features (line-spacing uniformity) for improved class separation
 - Cross-validation for more rigorous parameter tuning
 - Final presentation and polished webpage
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2. Current Results

Training Set Accuracy: 63.1% (41/65)

This was achieved through iterative improvements:

Version	Accuracy	Key Change
v1: Baseline	26.2%	HSV masking + hardcoded thresholds

v2: Multi-feature	47.7%	Nearest-centroid with 5 features
v3: Auto masking	50.8%	Auto-segmentation with fallback
v4: Distance metric	52.3%	Per-class feature-space distance
v5: Weighted	63.1%	Per-class distance weights

Confusion Matrix (v5)

	Pred: Optimal	Pred: Under	Pred: Over
True: Optimal	21	0	3
True: Under	7	7	6
True: Over	8	0	13

Per-Class Metrics

Class	Precision	Recall	F1
Optimal	0.583	0.875	0.700
Under-extruded	1.000	0.350	0.519
Over-extruded	0.591	0.619	0.605

3. Difficulties Encountered

Dataset Collection (biggest blocker)

Finding suitable first-layer images was far more difficult than anticipated. Existing Kaggle datasets contained general 3D printing images but not top-down first-layer views. We compiled 70+ YouTube timelapse links and 5 Kaggle datasets, then manually reviewed and extracted frames. Even after curation, 16 out of 81 images had to be set aside as questionable (wrong angle, multi-layer prints, text overlays, etc.), leaving 65 usable images - near the low end of our 50-75 target.

ROI Masking

The initial HSV thresholding with hardcoded ranges assumed dark build plates and failed on many images (textured PEI sheets, colored beds, reflective surfaces). Implementing automatic thresholding and a multi-strategy auto-masking approach significantly improved robustness, but masking quality still varies across the diverse dataset.

Feature Separability

The extracted features overlap substantially between classes. Under-extruded and over-extruded images have nearly identical mean gradient statistics (38.7 vs 38.4), making them hard to distinguish with global features. The current 5-feature vector captures general image statistics rather than the specific physical property we care about (extrusion line spacing and regularity). This is the core limitation.

Performance

Python implementations of morphological operations and edge detection were initially very slow (~30 s per 1080p image). We mitigated this by downscaling images to 512 px max and replacing loop-based dilation/erosion with vectorized stride-trick operations, reducing per-image time to ~1 s.

4. Changes from Proposal

Approach Modifications

- **Classifier:** The proposal described threshold-based classification using fill density ratios. We evolved this substantially - from simple thresholds to nearest-centroid to a feature-space distance classifier, as the original approach only achieved 26% accuracy.
- **Segmentation:** We added automatic image segmentation as the primary strategy, replacing the original HSV-only approach.
- **Image preprocessing:** Added a resize step not mentioned in the proposal, necessary to make the algorithms tractable.

Timeline Adjustments

- Classification work started early (overlapping with edge detection) because we needed to test the pipeline end-to-end
- The core pipeline is functional, but accuracy improvements remain the primary focus

Scope

The proposal's homography/perspective correction has not been implemented. Most curated images are already approximately top-down, and statistical features are largely invariant to mild perspective distortion, so this is deprioritized in favor of improving feature discriminability.

5. Plan for Remaining Work

1. **Held-out validation** - collect 15-20 new images, evaluate pipeline without retraining to get honest accuracy
2. **Line-spacing uniformity feature** - measure extrusion line periodicity directly via intensity profiles, targeting the core physical difference between defect types
3. **Cross-validation** - k-fold CV on training data for more rigorous parameter tuning
4. **Project webpage** - polish and finalize with validation results